MongoDB Exercise: Querying the Movies Database

Objective:
In this exercise, you'll work with a movies database in MongoDB to enhance your querying skills. You will upload a dataset, generate five unique queries, and analyze the results. You may create these queries manually or use MongoDB's AI module for assistance. After executing each query, export the results as a CSV file, assess whether the output matches your intended query purpose, and provide a detailed explanation.

Instructions:

1. **Upload the Movies Database:**
   - Import the movies dataset into MongoDB. Use a dataset provided by your instructor or find one online.
   - Verify the import by ensuring you can access a collection named `movies` in MongoDB.
2. **Generate 5 Queries:**
   - Create 5 unique queries to analyze specific aspects of the data. Some examples include:
     - Finding movies by genre or release years.
     - Filtering by ratings, directors, or other fields.
     - Analyzing trends, such as movies with specific actors, production studios, or popular keywords in their titles.
     - After running each query, evaluate whether it produced the intended results. Confirm that the output matches your specified genre, time period, or other criteria. If there are discrepancies, refine the query and note any adjustments.
3. **For Each Query, Provide:**
   - **Query Code:** Include the MongoDB query syntax used.
   - **Query Results:** Export the output of each query as a CSV file, showing at least the first 10 rows.
   - **Explanation and Analysis:**
     - *Purpose:* Describe what the query aims to retrieve or analyze.
     - *Structure:* Explain how the query is structured and why specific parameters were used.
     - *Assessment of Results:* Reflect on whether the output aligned with your intent and any adjustments made for accuracy.
4. **Submission Requirements:**
   - For each query, provide:
     - The query code
     - Query results in CSV format
     - A document with detailed explanations, including the purpose, structure, and assessment for each query
   - Upload all CSV files and the explanatory document to the assignment portal.

Example Query Outline:

1. **Query Code:**

```
db.movies.find({ "genre": "Action", "year": { "$gte": 2010 } })
```

2. **Query Results (CSV):**
   - Export the query output as a CSV file, showing at least the first 10 rows.
3. **Explanation and Assessment:**
   - *Purpose:* This query retrieves Action genre movies released from 2010 onward.
   - *Structure:* The query specifies `"genre": "Action"` to filter by genre and `"year": { "$gte": 2010 }` to include only movies from 2010 and beyond.
   - *Assessment:* The results aligned with the intended goal of retrieving post-2010 Action movies, with no adjustments needed.